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Inventor(s)	Watanabe; Shinya

Semiconductor device

Abstract

A semiconductor device according to the present embodiment includes a substrate having a first semiconductor circuit provided thereon. First pads are located on the substrate. A first insulating layer is located on an outer side of each of the first pads. Second pads are respectively bonded to the first pads. A second insulating layer is located on an outer side of each of the second pads and is bonded to the first insulating layer. The first pads each include a first conductive material, and a first insulating material located on an inner side of the first conductive material on a bonding surface of the first pads and the second pads.

Inventors: **Watanabe; Shinya (Yokkaichi, JP)**

Applicant: **Kioxia Corporation** (Tokyo, JP)

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Assignee: **Kioxia Corporation (Tokyo, JP)**

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Primary Examiner: Page; Dale E

Assistant Examiner: Baptiste; Wilner Jean

Attorney, Agent or Firm: Oblon, McClelland, Maier & Neustadt, L.L.P.

Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS
(1) This application is based upon and claims the benefit of priority from the prior Japanese Patent Application No. 2021-086411, filed on May 21, 2021, the entire contents of which are incorporated herein by reference.
FIELD
(2) The embodiments of the present invention relate to a semiconductor device.
BACKGROUND
(3) In recent years, a technology of sticking a plurality of semiconductor chips to electrically bond pads to each other is developed. Meanwhile, in a polishing method such as a CMP (Chemical Mechanical Polishing) method, dishing (a recess) sometimes occurs due to a difference in the quality of materials to be polished. If the pads on the bonding surface are recessed due to dishing, the contact resistance between the pads may be increased or an open circuit failure between the pads may occur when a plurality of semiconductor chips are stuck to each other.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) FIG. **1** is a sectional view illustrating a configuration example of a semiconductor package according to a first embodiment;
(2) FIG. **2** is a sectional view illustrating a configuration example of a portion of the semiconductor package according to the first embodiment;
(3) FIG. **3**A is a plan view illustrating a configuration example of pads;
(4) FIG. **3**B is a sectional view illustrating a configuration example of the pads;
(5) FIG. **3**C is a plan view illustrating one example of a configuration of a wiring layer;
(6) FIG. **4** is a sectional view illustrating a configuration example of a portion of a bonding surface;
(7) FIG. **5** is a sectional view illustrating one example of a manufacturing method of the pads according to the first embodiment;
(8) FIG. **6** is a sectional view illustrating one example of the manufacturing method of the pads in continuation from FIG. **5**;
(9) FIG. **7** is a sectional view illustrating one example of the manufacturing method of the pads in continuation from FIG. **6**;
(10) FIG. **8** is a sectional view illustrating one example of the manufacturing method of the pads in continuation from FIG. **7**;
(11) FIG. **9** is a sectional view illustrating one example of the manufacturing method of the pads in continuation from FIG. **8**;
(12) FIG. **10** is a sectional view illustrating one example of the manufacturing method of the pads in continuation from FIG. **9**;
(13) FIG. **11** is a sectional view illustrating one example of the manufacturing method of the pads

in continuation from FIG. **10**;

(14) FIG. **12** is a sectional view illustrating one example of a formation process of a region of through silicon vias of a circuit chip;

(15) FIG. **13** is a sectional view illustrating one example of the manufacturing method of the pads in continuation from FIG. **12**;

(16) FIG. **14** is a sectional view illustrating one example of the manufacturing method of the pads in continuation from FIG. **13**;

(17) FIG. **15** is a sectional view illustrating one example of the manufacturing method of the pads in continuation from FIG. **14**;

(18) FIG. **16** is a sectional view illustrating one example of the manufacturing method of the pads in continuation from FIG. **15**;

(19) FIG. **17** is a sectional view illustrating one example of the manufacturing method of the pads in continuation from FIG. **16**;

(20) FIG. **18**A is a sectional view illustrating another example of the manufacturing method of the pads;

(21) FIG. **18**B is a sectional view illustrating the manufacturing method of the pads in continuation from FIG. **18**A;

(22) FIG. **18**C is a sectional view illustrating the manufacturing method of the pads in continuation from FIG. **18**B;

(23) FIG. **18**D is a sectional view illustrating the manufacturing method of the pads in continuation from FIG. **18**C;

(24) FIG. **19**A is a sectional view illustrating still another example of the manufacturing method of the pads;

(25) FIG. **19**B is a sectional view illustrating the manufacturing method of the pads in continuation from FIG. **19**A;

(26) FIG. **19**C is a sectional view illustrating the manufacturing method of the pads in continuation from FIG. **19**B;

(27) FIG. **19**D is a sectional view illustrating the manufacturing method of the pads in continuation from FIG. **19**C;

(28) FIG. **19**E is a sectional view illustrating the manufacturing method of the pads in continuation from FIG. **19**D;

(29) FIG. **19**F is a sectional view illustrating the manufacturing method of the pads in continuation from FIG. **19**E;

(30) FIG. **19**G is a sectional view illustrating the manufacturing method of the pads in continuation from FIG. **19**F;

(31) FIG. **20**A is a sectional view illustrating another example of the formation process of the region of the through silicon vias of the circuit chip;

(32) FIG. **20**B is a sectional view illustrating another example of the formation process of the region of the through silicon vias in continuation from FIG. **20**A;

(33) FIG. **20**C is a sectional view illustrating another example of the formation process of the region of the through silicon vias in continuation from FIG. **20**B;

(34) FIG. **20**D is a sectional view illustrating another example of the formation process of the region of the through silicon vias in continuation from FIG. **20**C;

(35) FIG. **20**E is a sectional view illustrating another example of the formation process of the region of the through silicon vias in continuation from FIG. **20**D;

(36) FIG. **20**F is a sectional view illustrating another example of the formation process of the region of the through silicon vias in continuation from FIG. **20**E;

(37) FIG. **21** is a plan view illustrating a configuration example of pads according to a second embodiment;

(38) FIG. **22** is a sectional view illustrating a configuration example of a region of a bonding

surface according to the second embodiment;
(39) FIG. **23** is a plan view illustrating a configuration example of pads according to a third embodiment;
(40) FIG. **24** is a plan view illustrating a configuration example of pads according to a fourth embodiment; and
(41) FIG. **25** is a sectional view illustrating a configuration example of a region of a bonding surface according to a fifth embodiment.

DETAILED DESCRIPTION

(42) Embodiments will now be explained with reference to the accompanying drawings. The present invention is not limited to the embodiments. In the embodiments, the term "upper direction" or "lower direction" of a semiconductor chip occasionally differs from an upper direction or a lower direction based on a gravitational acceleration direction. In the present specification and the drawings, elements identical to those described in the foregoing drawings are denoted by like reference characters and detailed explanations thereof are omitted as appropriate.
(43) A semiconductor device according to the present embodiment includes a substrate having a first semiconductor circuit provided thereon. First pads are located on the substrate. A first insulating layer is located on an outer side of each of the first pads. Second pads are respectively bonded to the first pads. A second insulating layer is located on an outer side of each of the second pads and is bonded to the first insulating layer. The first pads each include a first conductive material, and a first insulating material located on an inner side of the first conductive material on a bonding surface of the first pads and the second pads.

First Embodiment

(44) FIG. **1** is a sectional view illustrating a configuration example of a semiconductor package **1** according to a first embodiment. The semiconductor package **1** of the present embodiment is an example of a package of semiconductor memories. However, the present embodiment can also be applied to other semiconductor devices.
(45) The semiconductor package **1** includes a wiring substrate **10**, metallic bumps **20**, solder balls **70**, a controller chip **30**, a memory chip stacked body **40** including a plurality of stacked memory chips, electrodes **50** provided to penetrate through the memory chips, and a sealing resin **60**.
(46) The wiring substrate **10** includes an insulator **11**, a wiring layer **12**, and a solder resist layer **13**. For example, an insulating material such as glass epoxy resin is used as the insulator **11**. The wiring layer **12** is a conductor provided on the front and back surfaces of the insulator **11**. For example, a low-resistance metallic material such as copper is used as the wiring layer **12**. The solder resist layer **13** is provided on the wiring layer **12**.
(47) The metallic bumps **20** are provided on the front surface side of the wiring substrate **10** and are electrically connected to portions of the wiring layer **12**, respectively. The solder balls **70** are provided on the back surface side of the wiring substrate **10** and are electrically connected to portions of the wiring layer **12**, respectively.
(48) The controller chip **30** is provided above the front surface of the wiring substrate **10**. The controller chip **30** is provided to control the memory chips.
(49) The memory chip stacked body **40** is stacked on the controller chip **30**. The memory chips are, for example, semiconductor chips on which NAND memory cells are mounted. The memory chips and the controller chip **30** are electrically connected via the electrodes **50**. The electrodes **50** transmit supply power, a ground voltage, a control signal, data, or the like. For example, a conductive material such as tungsten, nickel, copper, gold, aluminum, or polysilicon is used as the electrodes **50**.
(50) The sealing resin **60** is provided on the front surface of the wiring substrate **10** and seals the controller chip **30** and the memory chip stacked body **40**.
(51) FIG. **2** is a sectional view illustrating a configuration example of a portion of the semiconductor package **1** according to the first embodiment. FIG. **2** illustrates a cross section of

stacked two memory chips **40_1** and **40_2**. The memory chip **40_1** and the memory chip **40_2** are bonded to each other on a bonding surface B_chip.

(52) The memory chip **40_1** includes an array chip CH_A**1** including a memory cell array MCA**1**, and a circuit chip CH_C**1** including a CMOS (Complementary Metal Oxide Semiconductor) circuit CMOS**1**. The memory chip **40_2** includes an array chip CH_A**2** including a memory cell array MCA**2**, and a circuit chip CH_C**2** including a CMOS circuit CMOS**2**. In the memory chip **40_1**, one of the memory cell array MCA**1** and the CMOS circuit CMOS**1** may be a first semiconductor circuit and the other thereof may be a second semiconductor circuit. In the memory chip **40_2**, one of the memory cell array MCA**2** and the CMOS circuit CMOS**2** may be a first semiconductor circuit and the other thereof may be a second semiconductor circuit. The memory cell array MCA**1** and the CMOS circuit CMOS**1** included in the memory chip **40_1** may be a first semiconductor circuit and the memory cell array MCA**2** and the CMOS circuit CMOS**2** included in the memory chip **40_2** may be a second semiconductor circuit with the bonding surface B_chip interposed therebetween.

(53) Memory Chip **40_1**

(54) The array chip CH_A**1** includes the memory cell array MCA**1** covered with an interlayer dielectric film ILD**1_1**. The memory cell array MCA**1** includes a plurality of word lines WL**1** stacked in a Z direction and insulated from each other, and a plurality of columnar bodies CL**1** extending to penetrate through the stacked word lines WL**1** in the stacking direction (the Z direction). Memory cells MC**1** are provided to correspond to intersections between the word lines WL**1** and the columnar bodies CL**1**, respectively. One ends of the columnar bodies CL**1** are connected in common to a source line SL**1**. The other ends of the columnar bodies CL**1** are connected to any of bit lines BL**1** extending in a Y direction, respectively.

(55) The memory cell array MCA**1** is provided in an array region R_Arr. The word lines WL**1** extend in an X direction to a terrace region R_Trr and are formed in the manner of stairs in the terrace region R_Trr. A contact plug CC**1** is connected to a step surface of each of the word lines WL**1** formed in the stair manner. Each of the contact plugs CC**1** is electrically connected between an associated one of pads P**1_1***a* provided in the terrace region R_Trr and the associated word line WL**1** via a wiring layer W**1_1**. The pads P**1_1***a* are electrode pads exposed on a surface of the interlayer dielectric film ILD**1_1** and provided on a bonding surface B_mc**1** of the array chip CH_A**1**. The wiring layer W**1_1** electrically connects between the memory cell array MCA**1** and the pads P**1_1***a* via the contact plugs CC**1**, respectively.

(56) A peripheral region R_Pri is provided around the array region R_Arr and the terrace region R_Trr. The peripheral region R_Pri may be provided at various locations including a central portion of the memory chips as well as the peripheral portion of the memory chips. Contact plugs Cpri**1** are provided in the peripheral region R_Pri to penetrate through the interlayer dielectric film ILD**1_1** of the array chip CH_A**1** in the Z direction. One ends of the contact plugs Cpri**1** are electrically connected to the pads P**1_1***a* provided on the bonding surface B_mc**1** in the peripheral region R_Pri via the wiring layer W**1_1**, respectively. The other ends of the contact plugs Cpri**1** are electrically connected to pads P**1_1***b* provided on the opposite surface to the bonding surface B_mc**1** of the array chip CH_A**1**, respectively.

(57) The circuit chip CH_C**1** is provided below (in a −Z direction) the array chip CH_A**1** and includes the CMOS circuit CMOS**1** covered with an interlayer dielectric film ILD**1_2**. The CMOS circuit CMOS**1** is a circuit provided on a semiconductor layer SUB**1** and including a P-type MOSFET (MOS Field Effect Transistor) and an N-type MOSFET. The CMOS circuit CMOS**1** may include other semiconductor elements (for example, a resistive element or a capacitive element). The CMOS circuit CMOS**1** is covered with the interlayer dielectric film ILD**1_2**. A multilayer wiring layer W**1_2** is provided in the interlayer dielectric film ILD**1_2**. The multilayer wiring layer W**1_2** electrically connects between the CMOS circuit CMOS**1** and pads P**1_2***a*. The pads P**1_2***a* are electrode pads exposed on a surface of the interlayer dielectric film ILD**1_2** and provided on

the bonding surface B_mc**1** of the circuit chip CH_C**1**. The pads P**1_2***a* may be provided in any of the array region R_Arr, the terrace region R_Trr, and the peripheral region R_Pri.
(58) Through silicon vias TSV**1** are provided in the peripheral region R_Pri of the circuit chip CH_C**1**. The through silicon vias TSV**1** are portions of the electrodes **50**, respectively. The through silicon vias TSV**1** penetrate through the semiconductor layer SUB**1** in the Z direction and are electrically connected between associated ones of the pads P**1_2***a* and associated ones of pads P**1_2***b*, respectively. The pads P**1_2***b* are electrode pads provided at end portions of the through silicon vias TSV**1** on the opposite side to the bonding surface B_mc**1**, respectively.
(59) The array chip CH_A**1** and the circuit chip CH_C**1** are stuck to each other on the bonding surface B_mc**1**. The interlayer dielectric films ILD**1_1** and ILD**1_2** are bonded and the pads P**1_1***a* and P**1_2***a* are bonded, respectively, on the bonding surface B_mc**1**. This enables the CMOS circuit CMOS**1** of the circuit chip CH_C**1** to be electrically connected to the memory cell array MCA**1** via the multilayer wiring layer W**1_2**, the pads P**1_2***a* and P**1_1***a*, and the contact plugs CC**1**. As a result, the CMOS circuit CMOS**1** can control the memory cell array MCA**1**. The through silicon vias TSV**1** are electrically connected to the contact plugs Cpri**1** via the pads P**1_2***a* and P**1_1***a* and the wiring layer W**1_1**. The through silicon vias TSV**1** are provided, for example, to enable the supply power or the ground potential to be transmitted to chips in common.
(60) Memory Chip **40_2**
(61) The array chip CH_A**2** includes the memory cell array MCA**2** covered with an interlayer dielectric film ILD**2_1**. The memory cell array MCA**2** includes a plurality of word lines WL**2** stacked in the Z direction and insulated from each other, and a plurality of columnar bodies CL**2** extending to penetrate through the stacked word lines WL**2** in the stacking direction (the Z direction). Memory cells MC**2** are provided to correspond to intersections between the word lines WL**2** and the columnar bodies CL**2**, respectively. One ends of the columnar bodies CL**2** are connected in common to a source line SL**2**. The other ends of the columnar bodies CL**2** are connected to any of bit lines BL**2** extending in the Y direction, respectively.
(62) The memory cell array MCA**2** is provided in the array region R_Arr. The word lines WL**2** extend in the X direction to the terrace region R_Trr and are formed in the manner of stairs in the terrace region R_Trr. A contact plug CC**2** is connected to a step surface of each of the word lines WL**2** formed in the stair manner. Each of the contact plugs CC**2** is electrically connected between an associated one of pads P**2_1***a* provided in the terrace region R_Trr and the associated word line WL**2** via a wiring layer W**2_1**. The pads P**2_1***a* are electrode pads exposed on a surface of the interlayer dielectric film ILD**2_1** and provided on a bonding surface B_mc**2** of the array chip CH_A**2**. The wiring layer W**2_1** electrically connects between the memory cell array MCA**2** and the pads P**2_1***a* via the contact plugs CC**2**, respectively.
(63) The peripheral region R_Pri is provided around the array region R_Arr and the terrace region R_Trr. Contact plugs Cpri**2** are provided in the peripheral region R_Pri to penetrate through the interlayer dielectric film ILD**2_1** of the array chip CH_A**2** in the Z direction. One ends of the contact plugs Cpri**2** are electrically connected to the pads P**2_1***a* provided on the bonding surface B_mc**2** in the peripheral region R_Pri via the wiring layer W**2_1**, respectively. The other ends of the contact plugs Cpri**2** are electrically connected to pads P**2_1***b* provided on the opposite surface to the bonding surface B_mc**2** of the array chip CH_A**2**, respectively.
(64) The circuit chip CH_C**2** is provided below (in the −Z direction) the array chip CH_A**2** and includes the CMOS circuit CMOS**2** covered with an interlayer dielectric film ILD**2_2**. The CMOS circuit CMOS**2** is a circuit provided on a semiconductor layer SUB**2** and including a P-type MOSFET and an N-type MOSFET. The CMOS circuit CMOS**2** may include other semiconductor elements (for example, a resistive element or a capacitive element). The CMOS circuit CMOS**2** is covered with the interlayer dielectric film ILD**2_2**. A multilayer wiring layer W**2_2** is provided in the interlayer dielectric film ILD**2_2**. The multilayer wiring layer W**2_2** electrically connects between the CMOS circuit CMOS**2** and pads P**2_2***a*. The pads P**2_2***a* are electrode pads exposed

on a surface of the interlayer dielectric film ILD**2**_**2** and provided on the bonding surface B_mc**2** of the circuit chip CH_C**2**. The pads P**2**_**2***a* may be provided in any of the array region R_Arr, the terrace region R_Trr, and the peripheral region R_Pri.
(65) Through silicon vias TSV**2** are provided in the peripheral region R_Pri of the circuit chip CH_C**2**. The through silicon vias TSV**2** penetrate through the semiconductor layer SUB**2** in the Z direction and are electrically connected between associated ones of the pads P**2**_**2***a* and associated one of pads P**2**_**2***b*, respectively. The pads P**2**_**2***b* are electrode pads provided at end portions of the through silicon vias TSV**2** on the opposite side to the bonding surface B_mc**2**, respectively.
(66) The array chip CH_A**2** and the circuit chip CH_C**2** are stuck to each other on the bonding surface B_mc**2**. The interlayer dielectric films ILD**2**_**1** and ILD**2**_**2** are bonded and the pads P**2**_**1***a* and P**2**_**2***a* are bonded, respectively, on the bonding surface B_mc**2**. This enables the CMOS circuit CMOS**2** of the circuit chip CH_C**2** to be electrically connected to the memory cell array MCA**2** via the multilayer wiring layer W**2**_**2**, the pads P**2**_**2***a* and P**2**_**1***a*, and the contact plugs CC**2**. As a result, the CMOS circuit CMOS**2** can control the memory cell array MCA**2**. The through silicon vias TSV**2** are electrically connected to the contact plugs Cpri**2** via the pads P**2**_**2***a* and P**2**_**1***a* and the wiring layer W**2**_**1**. The through silicon vias TSV**2** are also provided, for example, to enable the supply power or the ground potential to be transmitted to chips in common.
(67) Bonding Between Memory Chips **40**_**1** And **40**_**2**
(68) The memory chip **40**_**1** and the memory chip **40**_**2** are bonded to each other on the bonding surface B_chip. The pads P**1**_**2***b* and the pads P**2**_**1***b* are respectively bonded to each other on the bonding surface B_chip. The memory chips **40**_**1** and **40**_**2** are electrically connected via the pads P**1**_**2***b* and P**2**_**1***b* bonded to each other. Accordingly, the through silicon vias TSV**1** and TSV**2** and the contact plugs Cpri**1** and Cpri**2** are electrically connected and can transmit, for example, the supply power or the ground potential to the stacked memory chips **40**_**1** and **40**_**2** in common.
(69) Configuration Of Pads P**1**_**1***a* And The Like
(70) FIG. **3**A is a plan view illustrating a configuration example of the pads P**1**_**1***a*. In FIG. **3**A, the pads P**1**_**1***a* are exposed from the surface of the interlayer dielectric film ILD**1**_**1** in a first planar view from a direction substantially perpendicular to the surface (the bonding surface B_mc**1**) of the interlayer dielectric film ILD**1**_**1** of the array chip CH_A**1** (a planar view seen from the Z direction). In the above planar view, each of the pads P**1**_**1***a* is surrounded by the interlayer dielectric film ILD**1**_**1** and has, for example, a substantially octagonal shape. The planar shape of each of the pads P**1**_**1***a* may be a shape of a polygon other than the octagon, a substantially circular shape, or a substantially elliptical shape.
(71) A barrier metal film **101**_**1***a*, a conductive material **102**_**1***a*, and an insulating material **103**_**1***a* are provided on the inner side of each of the pads P**1**_**1***a*.
(72) The barrier metal film **101**_**1***a* is provided at the outer edge of each of the pads P**1**_**1***a* and is located between the interlayer dielectric film ILD**1**_**1** or the insulating material **103**_**1***a* and the conductive material **102**_**1***a*. For example, a conductive material such as a stacked film including a titanium film and a titanium nitride film is used as the barrier metal film **101**_**1***a*.
(73) The conductive material **102**_**1***a* is provided on the inner side of each of the pads P**1**_**1***a* surrounded by the barrier metal film **101**_**1***a*. For example, a conductive material such as copper or tungsten is used as the conductive material **102**_**1***a*. The insulating material **103**_**1***a* is provided in the manner of islands on the inner side of the conductive material **102**_**1***a* and is surrounded by the conductive material **102**_**1***a*.
(74) In the above planar view, a plurality of the insulating materials **103**_**1***a* extend in the Y direction and each have an elongated shape on a surface of the conductive material **102**_**1***a*. In the above planar view, the insulating materials **103**_**1***a* are arrayed in the X direction orthogonal to the Y direction in the manner of stripes or the manner of lines and spaces on the surface of the conductive material **102**_**1***a*. In other words, the insulating materials **103**_**1***a* are provided in the manner of slits or strips extending substantially in parallel to each other. The insulating materials

103_1*a* are provided on the inner side of each of the pads P**1_1***a* and do not reach the barrier metal film **101_1***a* and the interlayer dielectric film ILD**1_1** in the above planar view. The insulating materials **103_1***a* may connect to the interlayer dielectric film ILD**1_1** below the pads P**1_1***a*. The same material (for example, a silicon dioxide film) as that of the interlayer dielectric film ILD**1_1** can be used as the insulating materials **103_1***a*.
(75) In the above planar view, the area of the insulating materials **103_1***a* in each of the pads P**1_1***a* is smaller than the area of the conductive material **102_1***a*. By setting the area of the conductive material **102_1***a* to be relatively large, the contact area with the conductive material **102_2***a* in the pads P**1_2***a* of the circuit chip CH_C**1** becomes larger and the contact resistance between each of the pads P**1_1***a* and the associated one of the pads P**1_2***a* can be suppressed to be low.
(76) The insulating materials **103_1***a* are formed of a material lower in the etching rate in the CMP process than the material (for example, a metallic material such as copper or tungsten) of the conductive material **102_1***a* (for example, an oxide film such as a silicon dioxide film, a nitride film such as a silicon nitride film, a carbide film such as a silicon carbide film, or a composite material thereof can be used as the insulating materials **103_1***a*). For example, the insulating materials **103_1***a* can be formed of a material physically harder and less likely to be polished than the material of the conductive material **102_1***a*. Alternatively, the insulating materials **103_1***a* may be formed of a material less likely to be chemically etched by an abrasive (slurry) than the material of the conductive material **102_1***a*. Therefore, in the CMP process, the insulating materials **103_1***a* serve as supporting posts on the inner side of the conductive material **102_1***a* and can reduce thinning of the film thickness at a central portion of the conductive material **102_1***a* and reduce production of a recess (dishing).
(77) A width Wp**1_1***a* in the X direction or the Y direction of each of the pads P**1_1***a* is, for example, about 1 micrometer (μm). A width W**103_1***a* of each of the insulating materials **103_1***a* is, for example, about several tens of nanometers.
(78) The wiring layer W**1_1** indicated by a broken line is provided below each of the pads P**1_1***a*. The wiring layer W**1_1** is electrically connected to the associated pad P**1_1***a* through via contacts V**1_1**. In the present embodiment, nine via contacts V**1_1** are provided between each of the pads P**1_1***a* and the associated wiring layer W**1_1**. However, the number of the via contacts V**1_1** is not limited to nine and can be any value. FIG. **3**C is a plan view illustrating one example of the configuration of the wiring layer W**1_1**. The wiring layer W**1_1** is formed in the shape of a cross within a substantially square frame below the associated pad P**1_1***a* in the planar view described above. The nine via contacts V**1_1** are provided on the wiring layer W**1_1**. The wiring layer W**1_1** may have a solid shape instead of the cross shape.
(79) FIG. **3**B is a sectional view illustrating a configuration example of the pads P**1_1**. FIG. **3**B illustrates a cross section along a line B-B in FIG. **3**A. The pads P**1_1***a* are embedded in the interlayer dielectric film ILD**1_1** and are exposed on the surface of the interlayer dielectric film ILD**1_1**. The conductive material **102_1***a* is electrically connected to the associated wiring layer W**1_1** provided therebelow through the via contacts V**1_1**.
(80) The insulating materials **103_1***a* can be parts of the interlayer dielectric film ILD**1_1** and can be of the same material. The height of the conductive material **102_1***a* is, for example, about 1 μm.
(81) As described above, according to the present embodiment, each of the pads P**1_1***a* includes the insulating materials **103_1***a* provided in the manner of islands on the inner side of the conductive material **102_1***a* in the planar view from the direction substantially perpendicular to the bonding surface B_mc**1**. The insulating materials **103_1***a* are formed of a material lower in the etching rate than the conductive material **102_1***a*.
(82) Therefore, in the CMP process to polish the interlayer dielectric film ILD**1_1** and the conductive material **102_1***a*, the insulating materials **103_1***a* serve as supporting posts within the conductive material **102_1***a* and can reduce dishing of the conductive material **102_1***a*.
(83) If the insulating materials **103_1***a* are not provided, the conductive material **102_1***a* is polished

in a relatively wide area. In this case, the inner side of the conductive material **102_1**_a_ is greatly dished and recessed.
(84) In contrast thereto, according to the present embodiment, the insulating materials **103_1**_a_ divide the conductive material **102_1**_a_ into relatively small areas and serve as supporting posts within the conductive material **102_1**_a_. Accordingly, dishing is suppressed on the inner side of the conductive material **102_1**_a_.
(85) It is preferable that the insulating materials **103_1**_a_ are arranged substantially uniformly in the conductive material **102_1**_a_. This can suppress dishing of the conductive material **102_1**_a_ from occurring locally greatly.
(86) While the pads P**1_1**_a_ have been explained with reference to FIGS. **3**A and **3**B, the pads P**1_2**_a_, P**2_1**_a_, P**2_2**_a_, P**1_2**_b_, and P**2_1**_b_ can be similarly configured. Therefore, dishing is suppressed in the CMP process also in pads P**1_2**_a_, P**2_1**_a_, P**2_2**_a_, P**1_2**_b_, and P**2_1**_b_ other than the pads P**1_1**_a_. Since the configurations of the pads P**1_2**_a_, P**2_1**_a_, P**2_2**_a_, P**1_2**_b_, and P**2_1**_b_ can be easily understood with reference to FIGS. **3**A and **3**B, detailed explanations thereof are omitted.
(87) FIG. **4** is a sectional view illustrating a configuration example of a portion of the bonding surface B_mc**1**. The pads P**1_1**_a_ on the side of the array chip CH_A**1** and the pads P**1_2**_a_ on the side of the circuit chip CH_C**1** are respectively bonded to each other on the bonding surface B_mc**1**.
(88) The pads P**1_1**_a_ and the pads P**1_2**_a_ both have the configuration illustrated in FIGS. **3**A and **3**B. Therefore, similarly to the pads P**1_1**_a_ illustrated in FIG. **3**A, the barrier metal film **101_2**_a_, the conductive material **102_2**_a_, and the insulating material **103_2**_a_ are provided on the inner side of each of the pads P**1_2**_a_ in a planar view from a direction substantially perpendicular to the surface of the interlayer dielectric film ILD**1_2**.
(89) The pads P**1_1**_a_, the interlayer dielectric film ILD**1_1**, the barrier metal film **101_1**_a_, the conductive material **102_1**_a_, and the insulating material **103_1**_a_ in FIGS. **3**A and **3**B are respectively read as the pads P**1_2**_a_, the interlayer dielectric film ILD**1_2**, the barrier metal film **101_2**_a_, the conductive material **102_2**_a_, and the insulating material **103_2**_a_ in the following explanations of the pads P**1_2**_a_.
(90) The barrier metal film **101_2**_a_ is provided at the outer edge of each of the pads P**1_2**_a_ and is located between the interlayer dielectric film ILD**1_2** or the insulating material **103_2**_a_ and the conductive material **102_2**_a_. For example, a conductive material such as a stacked film including a titanium film and a titanium nitride film is used as the barrier metal film **101_2**_a_.
(91) The conductive material **102_2**_a_ is provided on the inner side of each of the pads P**1_2**_a_ surrounded by the barrier metal film **101_2**_a_. For example, a conductive material such as copper or tungsten is used as the conductive material **102_2**_a_. The insulating material **103_2**_a_ is provided in the manner of islands on the inner side of the conductive material **102_2**_a_ and is surrounded by the conductive material **102_2**_a_.
(92) In the planar view described above, a plurality of the insulating materials **103_2**_a_ extend in the Y direction and each have an elongated shape on a surface of the conductive material **102_2**_a_. In the above planar view, the insulating materials **103_2**_a_ are arrayed in the X direction orthogonal to the Y direction in the manner of stripes or the manner of lines and spaces on the surface of the conductive material **102_2**_a_. In other words, the insulating materials **103_2**_a_ are provided in the manner of slits or strips extending substantially in parallel to each other. The insulating materials **103_1**_a_ are provided on the inner side of each of the pads P**1_2**_a_ and do not reach the barrier metal film **101_2**_a_ and the interlayer dielectric film ILD**1_2** in the above planar view. The same material (for example, a silicon dioxide film) as that of the interlayer dielectric film ILD**1_2** can be used as the insulating materials **103_2**_a_.
(93) The insulating materials **103_2**_a_ are formed of a material (for example, a silicon dioxide film) lower in the etching rate than the material (for example, a metallic material such as copper or tungsten) of the conductive material **102_2**_a_. For example, the insulating materials **103_2**_a_ may be

formed of a material physically harder and less likely to be polished than the material of the conductive material **102_2**_a_. Alternatively, the insulating materials **103_2**_a_ may be formed of a material less likely to be chemically etched by an abrasive (slurry) than the material of the conductive material **102_2**_a_. Therefore, in the CMP process, the insulating materials **103_2**_a_ serve as supporting posts on the inner side of the conductive material **102_2**_a_ and can reduce dishing of the conductive material **102_2**_a_.
(94) A width Wp**1_2**_a_ in the X direction or the Y direction of each of the pads P**1_2**_a_ is, for example, about 1 μm. A width W**103_2**_a_ of each of the insulating materials **103_2**_a_ is, for example, about several tens of nanometers.
(95) The pads P**1_2**_a_ are embedded in the interlayer dielectric film ILD**1_2** and are exposed on the surface of the interlayer dielectric film ILD**1_2**. The conductive material **102_2**_a_ is electrically connected to the associated wiring layer W**1_2** provided therebelow. The insulating materials **103_2**_a_ can be parts of the interlayer dielectric film ILD**1_2** and can be of the same material. The height of the conductive material **102_2**_a_ is, for example, about 1 μm.
(96) As described above, the pads P**1_1**_a_ and P**1_2**_a_ have substantially same configurations. Each of the pads P**1_1**_a_ and the associated pad P**1_2**_a_ are bonded to each other on the bonding surface B_mc**1** between the pads P**1_1**_a_ and the pads P**1_2**_a_ in such a manner that the extending direction of the insulating materials **103_1**_a_ and the extending direction of the insulating materials **103_2**_a_ are substantially same directions (for example, the Y direction). Accordingly, the conductive material **102_1**_a_ and the conductive material **102_2**_a_ are bonded to substantially face each other and match each other on the bonding surface B_mc**1** as illustrated in FIG. **4** when the array chip CH_A**1** is stuck to the circuit chip CH_C**1**. At that time, the pads P**1_1**_a_ and the pads P**1_2**_a_ are little dished and the conductive materials **102_1**_a_ and **102_2**_a_ are little recessed on the bonding surface B_mc**1**. That is, the conductive material **102_1**_a_ and **102_2**_a_ are provided to be substantially flush on the bonding surface B_mc**1**. Therefore, the conductive material **102_1**_a_ and the conductive material **102_2**_a_ can be bonded on the bonding surface B_mc**1** with a sufficiently low resistance while the insulating materials **103_1**_a_ and **103_2**_a_ are provided on the inner side of the conductive material **102_1**_a_ and the conductive material **102_2**_a_, respectively.
(97) If the insulating materials **103_1** and **103_2** are not provided, the conductive materials **102_1**_a_ and **102_2**_a_ are likely to be poorly bonded due to dishing in the CMP process while the areas of the conductive materials **102_1**_a_ and **102_2**_a_ on the bonding surface B_mc**1** are correspondingly increased. Therefore, there is a risk that the contact resistance between the conductive material **102_1**_a_ and the conductive material **102_2**_a_ is increased.
(98) In contrast thereto, according to the present embodiment, the insulating materials **103_1**_a_ and **103_2**_a_ are provided and therefore the areas of the conductive materials **102_1**_a_ and **102_2**_a_ on the bonding surface B_mc**1** are correspondingly decreased. However, dishing of the conductive materials **102_1**_a_ and **102_2**_a_ are suppressed and the conductive materials **102_1**_a_ and **102_2**_a_ are little recessed on the bonding surface B_mc**1**. Therefore, the contact resistance between the conductive material **102_1**_a_ and the conductive material **102_2**_a_ can be lowered and stabilized.
(99) A manufacturing method of the pads P**1_1**_a_ and P**1_2**_a_ according to the first embodiment is explained next.
(100) FIGS. **5** to **11** are sectional views illustrating one example of the manufacturing method of the pads P**1_1**_a_ according to the first embodiment. Since the manufacturing method of the pads P**1_2**_a_ is same as that of the pads P**1_1**_a_, detailed explanations thereof are omitted.
(101) First, the memory cell array MCA**1**, the interlayer dielectric film ILD**1_1**, and the like are formed on a substrate (for example, a silicon substrate) for the array chip CH_A**1**. Next, the wiring layer W**1_1** is formed in the interlayer dielectric film ILD**1_1** of the array chip CH_A**1**.
(102) Next, an insulating film is further deposited on the wiring layer W**1_1** and the interlayer dielectric film ILD**1_1**. The insulating film can be of the same material (for example, a silicon dioxide film) as that of the interlayer dielectric film ILD**1_1**. Therefore, the insulating film on the

wiring layer W**1_1** is also referred to as the interlayer dielectric film ILD**1_1**. The structure illustrated in FIG. **5** is thereby obtained.
(103) Next, the interlayer dielectric film ILD**1_1** on the wiring layer W**1_1** is processed using a lithography technique and an etching technique. Accordingly, the interlayer dielectric film ILD**1_1** on the wiring layer W**1_1** is processed into a pattern of the via contacts V**1_1** as illustrated in FIG. **6**.
(104) Subsequently, a barrier metal film **201_1***a* and a conductive material **202_1***a* are deposited on the interlayer dielectric film ILD**1_1** and the wiring layer W**1_1** as illustrated in FIG. **7**. For example, a stacked film including a titanium film and a titanium nitride film is used as the barrier metal film **201_1***a*. For example, a conductive material such as copper or tungsten is used as the conductive material **202_1***a*.
(105) Next, the barrier metal film **201_1***a* and the conductive material **202_1***a* are polished using the CMP method until the interlayer dielectric film ILD**1_1** is exposed. The via contacts V**1_1** each including the barrier metal film **201_1***a* and the conductive material **202_1***a* are thereby formed as illustrated in FIG. **8**.
(106) Next, an insulating film is further deposited on the via contacts V**1_1**. The insulating film can be of the same material (for example, a silicon dioxide film) as that of the interlayer dielectric film ILD**1_1**. Therefore, the insulating film on the via contacts V**1_1** is also referred to as the interlayer dielectric film ILD**1_1**. Subsequently, the interlayer dielectric film ILD**1_1** on the via contacts V**1_1** is processed into a pattern of the pads P**1_1***a* using a lithography technique and an etching technique as illustrated in FIG. **9**. A first concave portion Con_**1** may be formed on each of the pads P**1_1***a*, a first insulating layer may be formed around the first concave portion Con_**1**, and a first insulating material may be formed on the inner side of the first concave portion Con_**1**.
(107) Next, the barrier metal film **101_1***a* and the conductive material **102_1***a* are deposited on the interlayer dielectric film ILD**1_1** and the via contacts V**1_1** as illustrated in FIG. **10**. For example, a stacked film including a titanium film and a titanium nitride film is used as the barrier metal film **101_1***a*. For example, a conductive material such as copper or tungsten is used as the conductive material **102_1***a*.
(108) Subsequently, the barrier metal film **101_1***a* and the conductive material **102_1***a* are polished using the CMP method until the interlayer dielectric film ILD**1_1** is exposed. Accordingly, the pads P**1_1***a* each including the barrier metal film **101_1***a* and the conductive material **102_1***a* are formed as illustrated in FIG. **11**. The interlayer dielectric film ILD**1_1** exposed in the CMP process becomes the insulating materials **103_1***a* described above.
(109) The insulating materials **103_1***a* are provided on the inner side of the conductive material **102_1***a* in the manner of islands (for example, the manner of stripes or the manner of lines and spaces) as illustrated in FIG. **3**A. The insulating materials **103_1***a* function as supporting posts in the conductive material **102_1***a* in the CMP process for the barrier metal film **101_1***a* and the conductive material **102_1***a*. Accordingly, dishing (a recess) of the conductive material **102_1***a* in the pads P**1_1***a* is suppressed.
(110) The manufacturing method of the pads P**1_1***a* of the array chip CH_A**1** has been explained above. The pads P**1_2***a* of the circuit chip CH_C**1** are formed in the same manner as that of the pads P**1_1***a* while being connected to the CMOS circuit CMOS**1**. Therefore, dishing (a recess) of the conductive material **102_2***a* is suppressed also in the pads P**1_2***a*.
(111) Dishing of the pads P**1_1***a* of the array chip CH_A**1** and the pads P**1_2***a* of the circuit chip CH_C**1** is suppressed. Therefore, when the array chip CH_A**1** is stuck to the circuit chip CH_C**1**, the pads P**1_1***a* and the pads P**1_2***a* are sufficiently bonded to each other with almost no space therebetween as illustrated in FIG. **4**. As a result, an increase in the contact resistance of the pads between the array chip CH_A**1** and the circuit chip CH_C**1** can be suppressed to suppress an open circuit failure.
(112) While the bonding between the array chip CH_A**1** and the circuit chip CH_C**1** has been

explained above, the present embodiment can be applied also to bonding between the memory chips **40_1** and **40_2**.
(113) Bonding Between Memory Chips **40_1** And **40_2**
(114) As illustrated in FIG. **2**, the memory chip **40_1** and the memory chip **40_2** are bonded to each other on the bonding surface B_chip. The memory chip **40_1** and **40_2** have a same configuration.
(115) Each of the pads P**1_2***b* of the memory chip **40_1** and the associated one of the pads P**2_1***b* of the memory chip **40_2** are electrically connected to each other on the bonding surface B_chip. Each of the pads P**1_2***b* is electrically connected to the associated one of the through silicon vias TSV**1** provided in the circuit chip CH_C**1** of the memory chip **40_1** via a redistribution layer (not illustrated). Each of the pads P**2_1***b* is electrically connected to the associated one of the contact plugs Cpri**2** of the array chip CH_A**2** of the memory chip **40_2**.
(116) The pads P**1_2***b* and P**2_1***b* can have the same configuration as that of the pads P**1_1***a* illustrated in FIGS. **3**A and **3**B. Accordingly, each of the pads P**1_2***b* and the associated one of the pads P**2_1***b* are bonded in the same manner as the pads P**1_1***a* and the pads P**1_2***a* illustrated in FIG. **4**. Therefore, the effects of the present embodiment can be achieved also in the bonding between the memory chips **40_1** and **40_2**.
(117) FIGS. **12** to **17** are sectional views illustrating one example of a formation process of a region of the through silicon vias TSV**1** of the circuit chip CH_C**1**.
(118) First, the CMOS circuit CMOS**1** is formed on a substrate (for example, a silicon substrate) SUB**1** using a semiconductor manufacturing process. As illustrated in FIG. **12**, the CMOS circuit CMOS**1** is electrically connected to the through silicon vias TSV**1** via the pads P**1_2***a* and the wiring layer W**1_2** (or receiving electrodes for the through silicon vias), respectively. Illustrations of the CMOS circuit CMOS**1**, the pads P**1_2***a*, and the wiring layer W**1_2** are omitted in FIG. **13** and subsequent drawings.
(119) Next, holes are formed in the formation region of the through silicon vias TSV**1** using a lithography technique and an etching technique. A spacer dielectric film SP**1** is formed on the inner walls of the holes. Next, the material (for example, copper or tungsten) of the through silicon vias TSV**1** is embedded inside the spacer dielectric film (for example, a silicon dioxide film) SP**1** using a plating method or the like. Subsequently, the interlayer dielectric film ILD**1_2** is deposited on the substrate SUB**1**. The structure illustrated in FIG. **12** is thereby obtained.
(120) In this way, the through silicon vias TSV**1** are formed after the CMOS circuit is formed. Therefore, the through silicon vias TSV**1** are formed after high-temperature heat treatment of the CMOS circuit, which enables the material (for example, copper or tungsten) of the through silicon vias TSV**1** to be formed using the plating method. Ends of the through silicon vias TSV**1** on the side of the CMOS circuit can be electrically connected to the CMOS circuit or may be electrically connected to an external electrode.
(121) Next, the circuit chip CH_C**1** is stuck to the array chip CH_A**1**. At that time, the pads P**1_1***a* and the pads P**1_2***a* are respectively bonded to each other (see FIG. **2**).
(122) Next, the substrate SUB**1** is turned upside down as illustrated in FIG. **13**. Subsequently, the back side of the substrate SUB**1** is etched to expose ends of the through silicon vias TSV**1** and the spacer dielectric film SP**1** as illustrated in FIG. **14**.
(123) Next, insulating films **91** and **92** are deposited on the substrate SUB**1** and the through silicon vias TSV**1** as illustrated in FIG. **15**. The insulating film **91** is, for example, a silicon nitride film and the insulating film **92** is, for example, a silicon dioxide film.
(124) Next, the insulating films **91** and **92** are polished using the CMP method until the through silicon vias TSV**1** are exposed as illustrated in FIG. **16**. The through silicon vias TSV**1** are thereby formed in the substrate SUB**1**. The through silicon vias TSV**1** penetrate through the substrate SUB**1** in a state of being electrically insulated from the substrate SUB**1** by the spacer dielectric film SP**1**.
(125) Next, a redistribution layer RW**1** is formed as illustrated in FIG. **17**. Next, the pads P**1_2***b* are formed on the redistribution layer RW**1**. The configuration and formation method of the pads

P**1**_**2***b* are as explained with reference to FIGS. **3**A to **11**.

(126) Subsequently, the memory chips **40**_**1** and **40**_**2** are stuck to each other. The pads P**1**_**2***b* and the associated pads P**2**_**1***b* are thereby respectively stuck to each other as illustrated in FIG. **2**.

(127) In a case in which through silicon vias are provided in the array chip CH_A**1**, the through silicon vias in the array chip CH_A**1** can also be formed by the same method as illustrated in FIGS. **12** to **17**.

First Modification

(128) FIGS. **18**A to **18**D are sectional views illustrating another example of the manufacturing method of the pads P**1**_**1***a*. Since the manufacturing method of the pads P**1**_**2***a* is same as that of the pads P**1**_**1***a*, detailed explanations thereof are omitted.

(129) After the structure illustrated in FIG. **5** is formed, the interlayer dielectric film ILD**1**_**1** on the wiring layer W**1**_**1** is processed using a lithography technique and an etching technique. Accordingly, the interlayer dielectric film ILD**1**_**1** on the wiring layer W**1**_**1** is processed into the pattern of the via contacts V**1**_**1** as illustrated in FIG. **18**A.

(130) Next, the interlayer dielectric film ILD**1**_**1** is processed again using the lithography technique and the etching technique to process an upper portion of the interlayer dielectric film ILD**1**_**1** into the pattern of the pads P**1**_**1***a* as illustrated in FIG. **18**B. Accordingly, the pattern of the pads P**1**_**1***a* is formed in the upper portion of the interlayer dielectric film ILD**1**_**1** and the pattern of the via contacts V**1**_**1** is formed under the pattern of the pads P**1**_**1***a* to be continuous therewith.

(131) Subsequently, the barrier metal film **101**_**1***a* and the conductive material **102**_**1***a* are deposited on the interlayer dielectric film ILD**1**_**1** and the wiring layer W**1**_**1** as illustrated in FIG. **18**C.

(132) Next, the barrier metal film **101**_**1***a* and the conductive material **102**_**1***a* are polished using the CMP method until the interlayer dielectric film ILD**1**_**1** is exposed. Accordingly, the via contacts V**1**_**1** and the pads P**1**_**1***a* each including the barrier metal film **101**_**1***a* and the conductive material **102**_**1***a* are simultaneously formed as illustrated in FIG. **18**D.

(133) In this CMP process, the insulating materials **103**_**1***a* serve as supporting posts on the inner side of the conductive material **102**_**1***a* and can reduce dishing of the conductive material **102**_**1***a*.

(134) The via contacts V**1**_**1** and the pads P**1**_**1***a* are simultaneously formed in the first modification. Therefore, in the first modification, the pads P**1**_**1***a* can be formed in fewer processes than those in the first embodiment. The rest of the manufacturing process in the first modification can be same as that in the first embodiment. Accordingly, the first modification can achieve the effects identical to those of the first embodiment.

Second Modification

(135) FIGS. **19**A to **19**G are sectional views illustrating still another example of the manufacturing method of the pads P**1**_**1***a*. Since the manufacturing method of the pads P**1**_**2***a* is same as that of the pads P**1**_**1***a*, detailed explanations thereof are omitted.

(136) After the structure illustrated in FIG. **5** is formed, the interlayer dielectric film ILD**1**_**1** in the entire formation region of each of the pads P**1**_**1***a* in the interlayer dielectric film ILD**1**_**1** on the wiring layer W**1**_**1** is removed using a lithography technique and an etching technique. The structure illustrated in FIG. **19**A is thereby obtained.

(137) Next, the barrier metal film **101**_**1***a* and the conductive material **102**_**1***a* are deposited on the interlayer dielectric film ILD**1**_**1** and the wiring layer W**1**_**1** as illustrated in FIG. **19**B.

(138) Next, the barrier metal film **101**_**1***a* and the conductive material **102**_**1***a* are polished using the CMP method until the interlayer dielectric film ILD**1**_**1** is exposed. The barrier metal film **101**_**1***a* and the conductive material **102**_**1***a* are thereby formed in the entire formation region of each of the pads P**1**_**1***a* as illustrated in FIG. **19**C.

(139) Subsequently, an upper portion of the conductive material **102**_**1***a* is processed using a lithography technique and an etching technique to remove the conductive material **102**_**1***a* located in the formation region of the insulating materials **103**_**1***a*. The structure illustrated in FIG. **19**D is

thereby obtained.
(140) Next, a barrier metal film **101_3** is deposited on the interlayer dielectric film ILD**1_1** and the conductive material **102_1***a* as illustrated in FIG. **19**E.
(141) Next, the insulating material **103_1***a* is deposited on the barrier metal film **101_3** as illustrated in FIG. **19**F.
(142) Next, the insulating material **103_1***a* is polished using the CMP method until the interlayer dielectric film ILD**1_1** is exposed. The pads P**1_1***a* are thereby formed as illustrated in FIG. **19**G. Even when the conductive material **102_1***a* is exposed in this CMP process, the insulating materials **103_1***a* serve as supporting posts on the inner side of the conductive material **102_1***a* and dishing of the conductive material **102_1***a* can be reduced.
(143) In the second modification, the via contacts V**1_1** are formed in the entire formation region of each of the pads P**1_1***a*. In this case, the pads P**1_1***a* are connected to the wiring layer W**1_1** through the via contacts V**1_1**.
(144) Also in the second modification, the via contacts V**1_1** and the pads P**1_1***a* are simultaneously formed. Therefore, the pads P**1_1***a* can be formed in fewer processes in the second modification than in the first embodiment. The rest of the formation process of the second modification can be same as that in the first embodiment. Accordingly, the second modification can achieve the effects identical to those of the first embodiment. An embodiment of using the pads P**1_1***a* formed in the second modification will be described later with reference to FIG. **25**.
Third Modification
(145) FIGS. **20**A to **20**F are sectional views illustrating another example of the formation process of the region of the through silicon vias TSV**1** of the circuit chip CH_C**1**. In this modification, the through silicon vias TSV**1** are formed after the CMOS circuit is formed and the substrate SUB**1** is inverted.
(146) First, the CMOS circuit (not illustrated) is formed on the substrate SUB**1** and the interlayer dielectric film ILD**1_2** is deposited thereon. The structure illustrated in FIG. **20**A is thereby obtained.
(147) Next, holes are formed in the formation region of the through silicon vias TSV**1** using a lithography technique and an etching technique as illustrated in FIG. **20**B.
(148) Subsequently, as illustrated in FIG. **20**C, the spacer dielectric film SP**1** is formed on the inner wall of the holes and is etched back to remove the spacer dielectric film SP**1** on bottom portions of the holes.
(149) Next, the material of the through silicon vias TSV**1** is embedded on the inner side of the spacer dielectric film SP**1** using a plating method or the like as illustrated in FIG. **20**D.
(150) In this way, the through silicon vias TSV**1** are formed after formation of the CMOS circuit. Accordingly, the through silicon vias TSV**1** are formed after high-temperature heat treatment of the CMOS circuit and therefore the material (for example, copper or tungsten) of the through silicon vias TSV**1** can be formed into a film using the plating method.
(151) Next, the material of the through silicon vias TSV**1** is polished using the CMP method until the surface of the spacer dielectric film SP**1** is exposed. The through silicon vias TSV**1** are thereby formed in the substrate SUB**1** as illustrated in FIG. **20**E. The through silicon vias TSV**1** penetrate through the substrate SUB**1** in a state of being electrically insulated from the substrate SUB**1** by the spacer dielectric film SP**1**.
(152) Next, the redistribution layer RW**1** is formed as illustrated in FIG. **20**F. Subsequently, the pads P**1_2***b* are formed on the redistribution layer RW**1**. The configuration and formation method of the pads P**1_2***b* are as explained with reference to FIGS. **3**A to **11**.
(153) Next, the memory chips **40_1** and **40_2** are stuck to each other. Accordingly, the pads P**1_2***b* and the associated pads P**2_1***b* are respectively stuck to each other as illustrated in FIG. **2**.
(154) In a case in which through silicon vias are provided in the array chip CH_A**1**, the through silicon vias in the array chip CH_A**1** can also be formed in the same manner as that in the present

modification.

Second Embodiment

(155) FIG. **21** is a plan view illustrating a configuration example of the pads P**1_2***a* according to a second embodiment. In a case in which the pads P**1_1***a* and the pads P**1_2***a* have the same configuration as in the first embodiment, there is a possibility that the conductive material **102_1***a* faces the insulating materials **103_2***a* and the conductive material **102_2***a* faces the insulating materials **103_1***a* if the pads P**1_1***a* are relatively displaced from the associated pads P**1_2***a* in the X direction in FIG. **4**, respectively. In this case, there is a risk that the contact area between the conductive material **102_1***a* and the conductive material **102_2***a* becomes extremely small, which increases and destabilizes the contact resistance between the pads P**1_1***a* and the pads P**1_2***a*.

(156) In contrast thereto, the insulating materials **103_2***a* of the pads P**1_2***a* extend in a direction oblique to the X and Y directions in a planar view seen from the Z direction in the second embodiment. The configurations of the pads P**1_1***a* may be identical to those of the first embodiment.

(157) FIG. **22** is a sectional view illustrating a configuration example of a region of the bonding surface B_mc**1** according to the second embodiment. In the second embodiment, the pads P**1_1***a* and the pads P**1_2***a* are respectively bonded to each other in such a manner that the extending direction (for example, the Y direction) of the insulating materials **103_1***a* intersects with the extending direction (a direction oblique to the X and Y directions) of the insulating materials **103_2***a* on the bonding surface B_mc**1** when the array chip CH_A**1** and the circuit chip CH_C**1** are stuck to each other. A cross section along a line B-B in FIG. **21** is illustrated as the pad P**1_2***a* in FIG. **22**.

(158) As viewed from a direction perpendicular to the substrate, the insulating materials **103_1***a* separate from each other partially overlap with the conductive material **102_2***a*, and the insulating materials **103_2***a* separate from each other partially overlap with the conductive material **102_1***a*. Since the extending direction of the insulating materials **103_1***a* intersects with the extending direction of the insulating materials **103_2***a*, the contact area between the conductive material **102_1***a* and the conductive material **102_2***a* does not reduce so much even if the pads P**1_1***a* are displaced from the pads P**1_2***a* to some extent in the X or Y direction. Therefore, the second embodiment enables the contact resistance to be low and stable against the displacement between the pads P**1_1***a* and the pads P**1_2***a* on the bonding surface B_mc**1**.

Third Embodiment

(159) FIG. **23** is a plan view illustrating a configuration example of the pads P**1_1***a* according to a third embodiment. In the third embodiment, the conductive material **102_1***a* of each of the pads P**1_1***a* has a mesh structure including elongated shapes extending in the X direction and elongated shapes extending in the Y direction on the surface of the interlayer dielectric film ILD**1_1** in a planar view seen from the Z direction. Therefore, the insulating materials **103_1***a* are formed in the manner of islands (the manner of dots) and are arrayed two-dimensionally in a matrix in the X direction and the Y direction on the surface of the interlayer dielectric film ILD**1_1** in the planar view seen from the Z direction.

(160) Assuming one of the insulating materials **103_1***a* as a first insulating portion In**1_1**, an insulating portion arranged in the Y direction and closest to the first insulating portion In**1_1** is a second insulating portion In**2_1** and an insulating portion arranged in the X direction and closest to the first insulating portion In**1_1** is a third insulating portion In**3_1**.

(161) The insulating materials **103_1***a* are formed of a material (for example, a silicon dioxide film) lower in the etching rate than the material (for example, a metallic material such as copper or tungsten) of the conductive material **102_1***a*. For example, the insulating materials **103_1***a* can be formed of a material physically harder and less likely to be polished than the material of the conductive material **102_1***a*. Alternatively, the insulating materials **103_1***a* may be formed of a material less likely to be chemically etched by an abrasive (slurry) than the material of the

conductive material **102_1***a*. Therefore, in the CMP process, the insulating materials **103_1***a* serve as supporting posts on the inner side of the conductive material **102_1***a* and can reduce dishing of the conductive material **102_1***a*.

(162) The pads P**1_2***a* also have the same configuration as that of the pads P**1_1***a* in FIG. **23**. Accordingly, the conductive material **102_2***a* of each of the pads P**1_2***a* also has a mesh structure including elongated shapes extending in the X direction and elongated shapes extending in the Y direction on the surface of the interlayer dielectric film ILD**1_2** in a planar view seen from the Z direction although not illustrated. That is, in the planar view seen from the Z direction, the insulating materials **103_2***a* are formed in the manner of islands (the manner of dots) on the surface of the interlayer dielectric film ILD**1_2** and are arrayed two-dimensionally in a matrix in the X direction and the Y direction. Therefore, the insulating materials **103_2***a* serve as supporting posts on the inner side of the conductive material **102_2***a* in the CMP process and can reduce dishing of the conductive material **102_2***a*.

(163) This enables the contact resistance between the conductive material **102_1***a* and the conductive material **102_2***a* to be low and stable.

(164) The third embodiment may be combined with any one of the first embodiment, the second embodiment, the first modification, and the second modification. That is, the pads P**1_1***a* according to the third embodiment may be bonded to the pads P**1_2***a* according to any one of the first embodiment, the second embodiment, the first modification, and the second modification.

(165) The third embodiment may be used for bonding between the memory chips **40_1** and **40_2**. That is, the third embodiment may be applied to the pads P**1_2***b* of the memory chip **40_1** and the pads P**2_1***b* of the memory chip **40_2**. Accordingly, dishing of the pads P**1_2***b* and the pads P**2_1***b* is suppressed and the bonding between the memory chips **40_1** and **40_2** can also be stabilized with a low resistance.

Fourth Embodiment

(166) FIG. **24** is a plan view illustrating a configuration example of the pads P**1_2***a* according to a fourth embodiment. In the fourth embodiment, the conductive material **102_2***a* of each of the pads P**1_2***a* extends in a direction oblique to the X and Y directions in a planar view seen from the Z direction. The configurations of the pads P**1_1***a* may be identical to those of any one of the first to third embodiments and the first and second modifications. The distance between dot-like insulating materials closest in the X direction and the Y direction in each of the pads P**1_1***a* in the third embodiment is different from the distance between dot-like insulating materials closest in the X direction and Y direction when the conductive material **102_2***a* is oblique as in the fourth embodiment. For example, assuming one of the insulating materials **103_2***a* as a fourth insulating portion In**4_2**, an insulating portion arranged in the Y direction and closest to the fourth insulating portion In**4_2** is a fifth insulating portion In**5_2** and an insulating portion arranged in the X direction and closest to the fourth insulating portion In**4_2** is a sixth insulating portion In**6_2**. The distance between the first insulating portion In**1_1** and the second insulating portion In**2_1** as seen in the Y direction is shorter than the distance between the fourth insulating portion In**4_2** and the fifth insulating portion In**5_2**. The distance between the first insulating portion In**1_1** and the third insulating portion In**3_1** as seen in the X direction is shorter than the distance between the fourth insulating portion In**4_2** and the sixth insulating portion In**6_2**.

(167) In the fourth embodiment, the pads P**1_1***a* and the associated pads P**1_2***a* are respectively bonded to each other in such a manner that the extending direction of the conductive material **102_1***a* intersects with the extending direction of the conductive material **102_2***a* on the bonding surface B_mc**1** when the array chip CH_A**1** and the circuit chip CH_C**1** are stuck to each other. Since the extending direction of the insulating materials **103_1***a* intersects with the extending direction of the insulating materials **103_2***a*, the contact area between the conductive material **102_1***a* and the conductive material **102_2***a* does not change so much even if the pads P**1_1***a* are displaced to some extent in the X or Y direction from the pads P**1_2***a*. Therefore, the fourth

embodiment can stabilize the contact resistance against the displacement of the pads P**1**_**1***a* from the pads P**1**_**2***a* on the bonding surface B_mc**1**.
(168) Other configurations of the fourth embodiment may be identical to the corresponding ones of the first to third embodiments. Accordingly, the fourth embodiment can also achieve the effects of any one of the first to third embodiments.
Fifth Embodiment
(169) FIG. **25** is a sectional view illustrating a configuration example of the region of the bonding surface B_mc**1** according to a fifth embodiment. The pads P**1**_**1***a* and P**1**_**2***a* formed according to the second modification described above are used in the fifth embodiment.
(170) In the fifth embodiment, the via contacts V**1**_**1** are provided below each of the pads P**1**_**1***a* and are electrically connected in common to the conductive material **102**_**1***a*. The via contacts V**1**_**1** electrically connect the conductive material **102**_**1***a* to the wiring layer W**1**_**1**. In this way, the via contacts V**1**_**1** are provided to be integral with the conductive material **102**_**1***a* in the entire formation region of each of the pads P**1**_**1***a*.
(171) Accordingly, the conductive material **102**_**1***a* of the pads P**1**_**1***a* is slightly raised from the bonding surface B_mc**1** due to volume expansion (heat expansion) of the via contacts V**1**_**1** and the conductive material **102**_**1***a*.
(172) Also for the pads P**1**_**2***a*, the via contacts V**1**_**2** are similarly provided below each of the pads P**1**_**2** and are electrically connected in common to the conductive material **102**_**2***a*. The via contacts V**1**_**2** electrically connect the conductive material **102**_**2***a* to the wiring layer W**1**_**2**. In this way, the via contacts V**1**_**2** are also provided in the entire formation region of each of the pads P**1**_**2***a* to be integral with the conductive material **102**_**2***a*. Accordingly, the conductive material **102**_**2***a* of each of the pads P**1**_**2***a* is slightly raised from the bonding surface B_mc**1** due to volume expansion (heat expansion) of the via contacts V**1**_**2** and the conductive material **102**_**2***a*.
(173) With raising of the pads P**1**_**1***a* and P**1**_**2***a* from the bonding surface B_mc**1**, the pads P**1**_**1***a* and P**1**_**2***a* are respectively reliably bonded to each other on the bonding surface B_mc**1**. Therefore, the pads P**1**_**1***a* and P**1**_**2***a* can be connected to each other stably with a low resistance.
(174) While certain embodiments have been described, these embodiments have been presented by way of example only, and are not intended to limit the scope of the inventions. Indeed, the novel methods and systems described herein may be embodied in a variety of other forms; furthermore, various omissions, substitutions and changes in the form of the methods and systems described herein may be made without departing from the spirit of the inventions. The accompanying claims and their equivalents are intended to cover such forms or modifications as would fall within the scope and spirit of the inventions.

Claims

1. A semiconductor device comprising: a substrate having a first semiconductor circuit provided thereon; first pads located on the substrate; second pads respectively bonded to the first pads; a first insulating layer located outside of each of the first pads on a bonding surface of the first pads and the second pads; and a second insulating layer located outside of each of the second pads on the bonding surface and bonded to the first insulating layer, wherein the first pads each comprise a first area including a first conductor and a second area including a third insulating layer on the bonding surface, the second area is located inside of the first area on the bonding surface, the first semiconductor circuit comprises one of a memory cell array and a CMOS circuit configured to control the memory cell array, and the device further comprises a second semiconductor circuit comprising the other of the memory cell array and the CMOS circuit configured to control the memory cell array on an opposite side to the first semiconductor circuit with the bonding surface interposed between the first semiconductor circuit and the second semiconductor circuit.
2. The device of claim 1, wherein the second area comprises a first extending portion extending in

a first direction on the bonding surface.
3. The device of claim 1, wherein the second area is smaller than the first area on the bonding surface.
4. The device of claim 1, wherein the second pads each comprise a third area including a second conductor and a fourth area including a fourth insulating layer on the bonding surface, the fourth area is located inside of the third area on the bonding surface.
5. The device of claim 4, wherein the fourth area comprises a second extending portion extending in a second direction lying in an in-plane direction of the bonding surface and different from the first direction on the bonding surface.
6. The device of claim 1, wherein on the bonding surface, the third insulating layer comprises a first insulating portion, a second insulating portion arranged alongside of the first insulating portion in the first direction and being closest thereto, and a third insulating portion arranged alongside of the first insulating portion in a second direction lying in an in-plane direction of the bonding surface and perpendicular to the first direction, and being closest thereto.
7. The device of claim 6, wherein on the bonding surface, the fourth insulating layer comprises a fourth insulating portion, a fifth insulating portion arranged alongside of the fourth insulating portion in the first direction and closest thereto, and a sixth insulating portion arranged alongside of the fourth insulating portion in the second direction and closest thereto, and when a distance between the first insulating portion and the second insulating portion closest in the first direction is a first distance, and a distance between the fourth insulating portion and the fifth insulation portion closest in the first direction is a second distance, the first distance is different from the second distance.
8. The device of claim 1, wherein the first insulating layer and the third insulating layer are simultaneously formed.
9. The device of claim 1, wherein the first insulating layer and the first insulator comprise oxygen and silicon, and the third insulating layer comprises copper, gold, or copper and gold.
10. A semiconductor device comprising: a substrate having a first semiconductor circuit provided thereon; first pads located on the substrate; second pads respectively bonded to the first pads; a first insulating layer located outside of each of the first pads on a bonding surface of the first pads and the second pads; and a second insulating layer located outside of each of the second pads on the bonding surface and bonded to the first insulating layer, wherein the first pads each comprise a first area including a first conductor and a second area including a third insulating layer on the bonding surface, the second area is located inside of the first area on the bonding surface, the first semiconductor circuit comprises a first memory cell array and a first CMOS circuit configured to control the first memory cell array, and the device further comprises a second semiconductor circuit comprising a second memory cell array and a second CMOS circuit configured to control the second memory cell array on an opposite side to the first semiconductor circuit with the bonding surface interposed between the first semiconductor circuit and the second semiconductor circuit.
11. The device of claim 10, wherein the second area comprises a first extending portion extending in a first direction on the bonding surface.
12. The device of claim 10, wherein the second area is smaller than the first area on the bonding surface.
13. The device of claim 10, wherein the second pads each comprise a third area including a second conductor and a fourth area including a fourth insulating layer on the bonding surface, the fourth area is located inside of the third area on the bonding surface.
14. The device of claim 13, wherein the fourth area comprises a second extending portion extending in a second direction lying in an in-plane direction of the bonding surface and different from the first direction on the bonding surface.
15. The device of claim 10, wherein on the bonding surface, the third insulating layer comprises a first insulating portion, a second insulating portion arranged alongside of the first insulating portion

in the first direction and being closest thereto, and a third insulating portion arranged alongside of the first insulating portion in a second direction lying in an in-plane direction of the bonding surface and perpendicular to the first direction, and being closest thereto.

16. The device of claim 15, wherein on the bonding surface, the fourth insulating layer comprises a fourth insulating portion, a fifth insulating portion arranged alongside of the fourth insulating portion in the first direction and closest thereto, and a sixth insulating portion arranged alongside of the fourth insulating portion in the second direction and closest thereto, and when a distance between the first insulating portion and the second insulating portion closest in the first direction is a first distance, and a distance between the fourth insulating portion and the fifth insulation portion closest in the first direction is a second distance, the first distance is different from the second distance.

17. The device of claim 10, wherein the first insulating layer and the third insulating layer are simultaneously formed.

18. The device of claim 10, wherein the first insulating layer and the third insulating layer comprise oxygen and silicon, and the first conductor comprises copper, gold, or copper and gold.